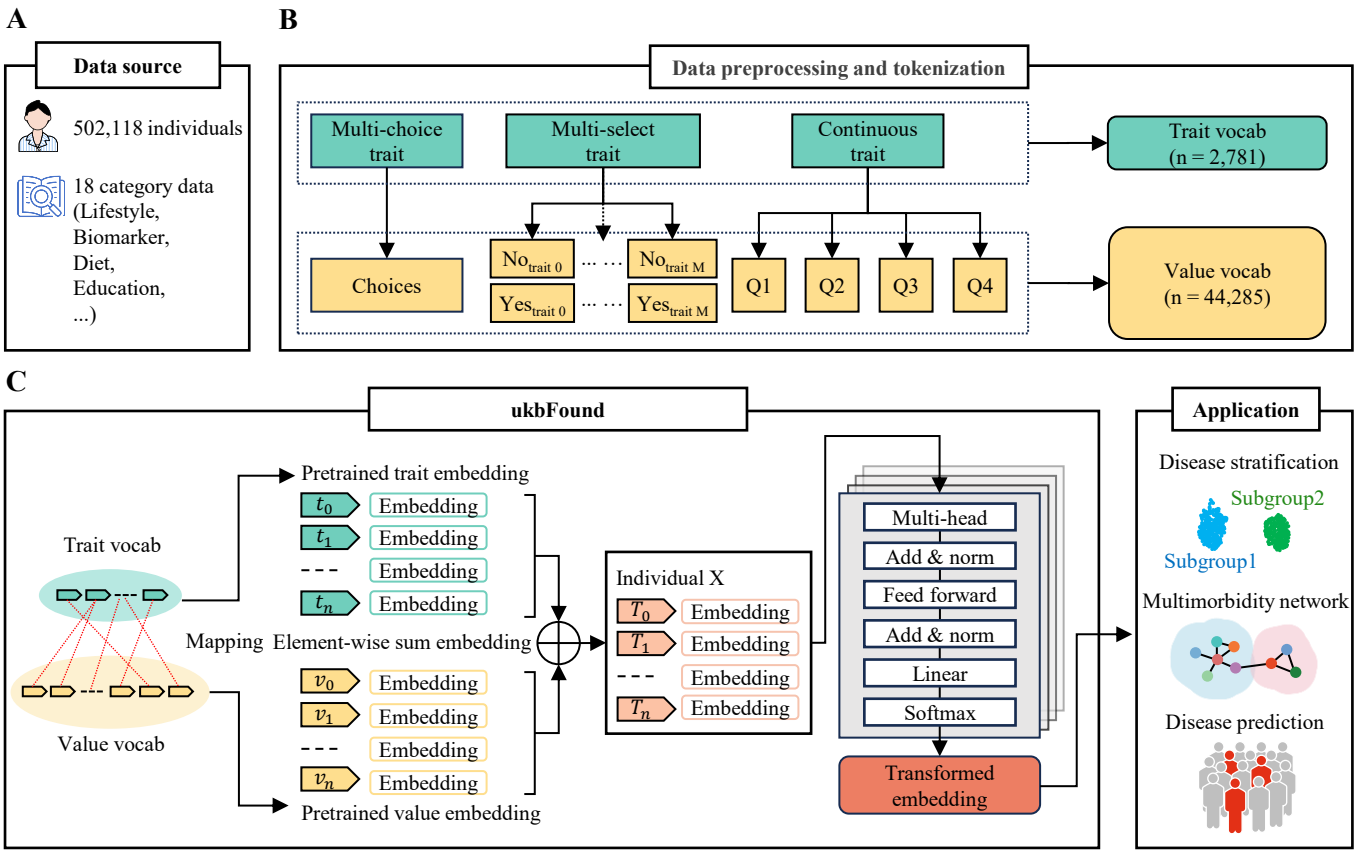

A
Data source
502,118 individuals
18 category data
(Lifestyle,
Biomarker,
Diet,
Education,
...)
B
Data preprocessing and tokenization
Multi-choice trait
Multi-select trait
Continuous trait
Trait vocab
(n = 2,781)
Choices
No trait 0
No trait M
Yes trait 0
Yes trait M
Q1
Q2
Q3
Q4
Value vocab
(n = 44,285)
C
ukbFound
Application
Trait vocab
Value vocab
Mapping
Pretrained trait embedding
Pretrained value embedding
Embedding
Element-wise sum embedding
Individual X
Multi-head
Add & norm
Feed forward
Add & norm
Linear
Softmax
Transformed embedding
Disease stratification
Subgroup2
Subgroup1
Multimorbidity network
Disease prediction